

Performance Analysis of IEEE 802.11e EDCF under Saturation Condition

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Abstract—This paper investigates Enhanced Distributed Coordination Function (EDCF) differentiation functionality and effectiveness in the emerging standard IEEE 802.11e via numerical and simulation results. An analytical model is proposed to derive saturation throughputs, saturation delays, and frame dropping probabilities for EDCF differentiation services. Effects of different parameters on the EDCF differentiation mechanism are obtained to provide some insights on the protocol.

I. INTRODUCTION

IEEE 802.11 medium access control (MAC) employs a mandatory contention-based channel access function called Distributed Coordination Function (DCF), and an optional centrally controlled channel access function called Point Coordination Function (PCF) [1]. The DCF adopts a carrier sense multiple access with collision avoidance (CSMA/CA) with binary exponential backoff. The current DCF is unsuitable for multimedia applications with Quality of Service (QoS) requirements.

To support MAC-level QoS, the IEEE 802.11 standardization committee is currently working on IEEE 802.11e [2], a supplement to the original IEEE 802.11 MAC. IEEE 802.11e will support multimedia applications such as voice and video over the IEEE 802.11 WLANs. The IEEE 802.11e MAC employs a channel access function called Hybrid Coordination Function (HCF), which includes a contention-based HCF part and a contention-free centrally controlled HCF part. The contention-based HCF part is also called Enhanced Distributed Coordination Function (EDCF). EDCF provides a priority scheme by differentiating the inter-frame space, the initial window size, and the maximum window size. Mangold and Choi et al. [3] and Choi et al. [4] introduced IEEE 802.11e EDCF and provided performance studies via simulations. Xiao [5] provided a good survey on the IEEE 802.11e standard. Xiao and Li [6] provided a performance analysis for the EDCF via simulations and effects of the buffer size on the EDCF queuing delay were studied. Xiao, Li, and Choi [7] proposed two-level QoS protection and guarantee mechanisms via simulations.

In the short paper [8], we proposed an analytical model (called the Old Model) for backoff-based priority schemes, based on Bianchi's model [9-10], and Ziouva and Antonakopoulos's model [11] (referred to as ZA's model). Both Bianchi's model and ZA's model are for the original DCF. In this paper, we propose a new improved analytical model, called the New Model. We provide following comparisons among Bianchi's model [9-10], ZA's model [11], the Old Model [8], and the New Model, as follows.

- ZA's model is based on Bianchi's model. Both the Old Model and the New Model are based on Bianchi's model and ZA's model. If assuming that there is only one class, the Old Model is equivalent to ZA's model.
- Both the Old Model and the New Model are for priorities,

whereas Bianchi's model and ZA's model are not.

- The New Model assumes that there is a finite retransmission limit (retry limit), whereas Bianchi's model, ZA's model, and the Old Model all assume an infinite retry limit. Therefore, the New Model is more consistent with the IEEE 802.11 standards [1]. Furthermore, the New Model can analyze a differentiation among classes with different retry limits. According to the author's knowledge, differentiating the retry limit is novel, and it is reasonable since for real-time multimedia traffic with sensitive delay requirements, retransmitted frames may be too late to be useful, and a smaller retry limit is appropriate, whereas some non-real-time transmissions may need a larger retry limit to enhance reliable transmissions. Furthermore, some new performance metrics such as the frame dropping probability can also be derived and studied. Note that paper [15] also considers retry limit for the original DCF.
- Bianchi's model assumes that the backoff counter is always decremented during a busy slot. This assumption is not consistent with the IEEE 802.11 standards since the backoff counter is stopped when a transmission is detected on the channel, and reactivated when the channel is sensed idle again [1]. ZA's model improves Bianchi's model by removing this assumption. Both the Old Model and the New Model follow ZA's model in this aspect, and therefore remove this assumption too.
- ZA's model and the Old Model assume that after every successful transmission, a station can transmit if the medium is idle without entering the backoff stage. Therefore, ZA's model introduces a non-backoff stage. This assumption is not consistent with the IEEE 802.11 standards. Bianchi's model has not such an assumption. The New Model has not such an assumption either.
- Another important contribution of this paper is to provide a new and accurate delay model other than ZA's model and the Old Model. The delay model is much simpler, more accurate, and easier to understand.
- Besides providing priority analysis, the New Model is an improved and better model when there is only one class than Bianchi's model and ZA's model. In other words, the New Model when there is only one class is a better model for the original DCF, especially for the delay model.

II. EDCF PRIORITY SCHEMES

The contention-based channel access function of IEEE 802.11e, EDCF, adopts eight different priorities, which are further mapped into four Access Categories (AC). Access Categories are achieved by differentiating the arbitration inter-frame space (AIFS), the initial window size, and the maximum window size. For the AC i ($i=0,1,2,3$) class, the initial

backoff window size is $CW_{\min}[i]$, the maximum backoff window size is $CW_{\max}[i]$, and the arbitration inter-frame space is $AIFS[i]$. Note that $CW_{\min}[i]$ is an EDCF notation, and is the same as $W_{i,0}$, a notation for our analytical model. For $0 \leq i < j \leq 3$, we have $CW_{\min}[i] \geq CW_{\min}[j]$, $CW_{\max}[i] \geq CW_{\max}[j]$ and $AIFS[i] \geq AIFS[j]$, and at least one of above inequalities must be “not equal to”. In other words, EDCF adopts $AIFS[i]$ ($i=0,1,\dots,3$), $CW_{\min}[i]$ ($i=0,1,\dots,3$), and $CW_{\max}[i]$ ($i=0,1,\dots,3$) instead of DIFS, $CW_{\min}$, and $CW_{\max}$, respectively. If one class has a smaller $AIFS$ or $CW_{\min}$ or $CW_{\max}$, the class’s traffic has a better chance to access the wireless medium earlier.

Four transmission queues are implemented in a station, and each queue supports one AC class, behaving roughly as a single DCF entity in the original IEEE 802.11 MAC. It is assumed that a payload from a higher layer is labeled with a priority value, and it is pushed into the corresponding queue with the same priority value. Each queue acts as an independent MAC entity and performs the same DCF function, with a different inter-frame space ($AIFS[i]$), a different initial window size ($CW_{\min}[i]$), and a different maximum window size ($CW_{\max}[i]$). Each queue has its own backoff counter ($BO[i]$), which acts independently the same way as the original DCF backoff counter introduced in the previous section. If there is more than one queue finishing the backoff at the same time, the highest priority frame is chosen to transmit by the virtual collision handler. Other lower priority frames whose backoff counters also reach zeros will increase their backoff counters with $CW_{\min}[i]$ ($i=0,1,\dots,3$), accordingly. Furthermore, we have $AIFS[i] > PIFS$, where $PIFS$ is point (coordination function) inter-frame space. Note that we only evaluate backoff-based part priority schemes of EDCF, but not IFS-based priority schemes in this paper. The relationships among $W_{i,j}$, $CW_{i,\max}$, $CW_{i,\min}$, $\sigma_i = 2$, and $L_{i,\text{retry}}$ are given as follows.

$$W_{i,j} = \begin{cases} \sigma_i^j W_{i,0} & \text{for } j=0,1,\dots,m_i-1, \text{ if } L_{i,\text{retry}} > m_i \\ \sigma_i^{m_i} W_{i,0} = CW_{i,\max} & \text{for } j=m_i,\dots,L_{i,\text{retry}}, \text{ if } L_{i,\text{retry}} > m_i \\ \sigma_i^j W_{i,0} & \text{for } j=0,1,\dots,L_{i,\text{retry}}, \text{ if } L_{i,\text{retry}} \leq m_i \end{cases} \quad (1)$$

where $m_i = \log_{\sigma_i}(CW_{i,\max}/CW_{i,\min})$ and $\sigma_i = 2$

III. ANALYTICALLY MODELING MULTIPLE PRIORITY CLASSES

Based on Bianchi’s model [9-10] and ZA’s model [11], an analytic model for multiple prioritized classes of traffic under high traffic condition is proposed. We assume that each station belongs to one and only one priority class (we will lose this assumption for EDCF in the next section), and always has frames ready to send. We assume that traffic is classified into N priority classes: $i=0,\dots,N-1$, and we can define priority schemes as follows.

A. An Analytical Model

For a given station in the priority i class ($i=0,\dots,N-1$), $b(i,t)$ is defined as a random process representing the value of the backoff counter at time t , and $s(i,t)$ is defined as the random process representing the backoff stage

j ($j=0,1,\dots,L_{i,\text{retry}}$) where $L_{i,\text{retry}}$ is the retry limit, for the priority i class. The value of the backoff counter $b(i,t)$ is uniformly chosen in the range $(0,1,\dots,W_{i,j}-1)$, where $W_{i,j} = \lceil (\sigma_i)^j W_{i,0} \rceil$. Let p_i denote the probability that a transmitted frame collides, and p_i also equals to the probability that a station in the backoff stage for the priority i class senses the channel busy. Similar to Bianchi’s model [9-10] and ZA’s model [11], the bi-dimensional random process $\{s(i,t), b(i,t)\}$ is discrete-time Markov chain. Therefore, the state of each station in the priority i class is described by $\{i, j, k\}$, where j stands for the backoff stage taking values from $(0,1,\dots,L_{i,\text{retry}})$, and k stands for the backoff delay taking values from $(0,1,\dots,W_{i,j}-1)$ in timeslots. The state transition diagram for the priority i class is shown in Fig. 1.

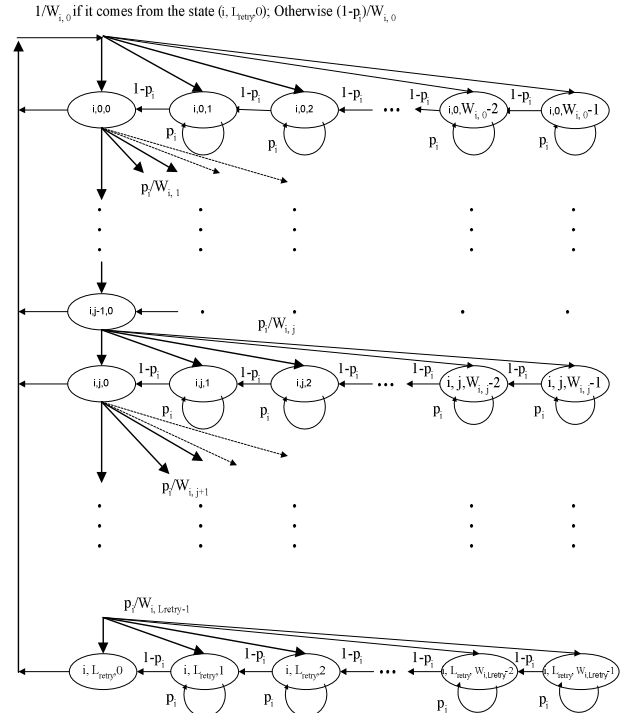


Fig. 1 The state transition diagram for the priority i class

Let $b_{i,j,k} = \lim_{t \rightarrow \infty} \Pr\{s(i,t) = j, b(i,t) = k\}$ be the stationary distribution of the Markov chain. In steady state, we have.

$$b_{i,j,0} = p_i^j b_{i,0,0} \quad 0 \leq j \leq L_{i,\text{retry}} \quad (2)$$

$$b_{i,j,k} = \frac{W_{i,j} - k}{W_{i,j}} \frac{1}{1 - p_{i,b}} b_{i,j,0} \quad 0 \leq j \leq L_{i,\text{retry}}, 1 \leq k \leq W_{i,j} - 1 \quad (3)$$

$$\sum_{j=0}^{L_{i,\text{retry}}} \sum_{k=0}^{W_{i,j}-1} b_{i,j,k} = 1 \quad (4)$$

From (2)-(4), we have,

$$b_{i,0,0} = \frac{1}{\sum_{j=0}^{L_{i,\text{retry}}} \left[1 + \frac{1}{1 - p_i} \sum_{k=1}^{W_{i,j}-1} \frac{W_{i,j} - k}{W_{i,j}} \right] p_i^j} \quad (5)$$

Let τ_i be the probability that a station in the priority i class transmits during a generic slot time. We have

$$\tau_i = \sum_{j=0}^{L_{i,\text{retry}}} b_{i,j,0} = b_{i,0,0} \frac{1 - p_i^{L_{i,\text{retry}}+1}}{1 - p_i} \quad (6)$$

Let n_i ($i=0,\dots,N-1$) denote the number of stations in the

priority i class. Let p_b denote the probability that the channel is busy. We have

$$p_i = 1 - \left(\prod_{h=0}^{i-1} (1 - \tau_h)^{n_h} \right) (1 - \tau_i)^{n_i-1} \left(\prod_{h=i+1}^{N-1} (1 - \tau_h)^{n_h} \right) \quad (7)$$

$$p_b = 1 - \prod_{h=0}^{N-1} (1 - \tau_h)^{n_h} \quad (8)$$

B. Saturation Throughput

Let $p_{s,i} (i=0, \dots, N-1)$ denote the probability that a successful transmission occurs in a slot time for the priority i class, and let p_s denote the probability that a successful transmission occurs in a slot time. We have

$$p_{s,i} = n_i \tau_i (1 - \tau_i)^{n_i-1} \prod_{h=0, h \neq i}^{N-1} (1 - \tau_h)^{n_h} \quad (9)$$

$$p_s = \sum_{i=0}^{N-1} p_{s,i} = \sum_{i=0}^{N-1} \frac{n_i \tau_i}{1 - \tau_i} (1 - p_b) \quad (10)$$

Let $S_i (i=0, \dots, N-1)$ denote the normalized throughput for the priority i class. Let δ , $T_{E(L)}$, T_s , and T_c denote the duration of an empty slot time, the time to transmit the average payload, the average time that the channel is sensed busy because of a successful transmission, and the average time that the channel has a collision, respectively. We have,

$$S_i = \frac{E(\text{payload transmission time in a slot time for the } i \text{ class})}{E(\text{length of a slot time})} \quad (11)$$

$$= \frac{p_{s,i} T_{E(L)}}{(1 - p_b) \delta + p_s T_s + [p_b - p_s] T_c}$$

Let T_H , T_{ACK} , $SIFS$, L^* , $T_{E(L^*)}$, and γ denote the time to transmit the header (including MAC header, PHY header, and/or tail), the time to transmit an ACK, SIFS time, the length of the longest frame in a collision, the time to transmit a payload with length $E(L^*)$, and the time of the propagation delay, respectively. For the basics access method, we have,

$$T_s^{basic} = T_H + T_{E(L)} + SIFS + \gamma + T_{ACK} + DIFS + \gamma \quad (12)$$

$$T_c^{basic} = T_H + T_{E(L^*)} + DIFS + \gamma \quad (13)$$

Let T_{RTS} and T_{CTS} denote the time to transmit an RCS frame and the time to transmit a CTS frame, respectively. For the RTS/CTS access model, we have,

$$T_s^{rts/cts} = T_{RTS} + SIFS + \gamma + T_{CTS} + SIFS + \gamma + T_H + T_{E(L)} + SIFS + \gamma + T_{ACK} + DIFS + \gamma \quad (14)$$

$$T_c^{rts/cts} = T_{RTS} + DIFS + \gamma \quad (15)$$

C. Frame Dropping Probability

Let $P_{i,drop} (i=0, \dots, N-1)$ denote the frame-dropping probability for the priority i class. Let $P_{i,success} (i=0, \dots, N-1)$ denote the frame-successful probability for the priority i class.

$$P_{i,drop} = p_i^{L_{i, retry}+1} \quad (16)$$

$$P_{i,success} = \sum_{j=0}^{L_{i, retry}} p_i^j (1 - p_i) = (1 - p_i) \frac{(1 - p_i^{L_{i, retry}+1})}{(1 - p_i)} = 1 - p_i^{L_{i, retry}+1} \quad (17)$$

D. Saturation Delay

Saturation delay is the average delay under the saturation condition, and it includes the medium access delay (due to backoff, collisions, etc.), transmission delay, and inter-frame spaces (such as SIFS). Let $X_i (i=0, \dots, N-1)$ denote the random

variable representing the total number of backoff slots, which a frame encounters without considering the case when the counter freezes, for the priority i class. Let $B_i (i=0, \dots, N-1)$ denote the random variable representing the total number of slots when the counter freezes, which a frame encounters, for the priority i class. Let $E(N_{i, retry})$ denote the average number of retries for the priority i class. We have,

$$E(X_i) = \sum_{j=0}^{L_{i, retry}} \frac{p_i^j (1 - p_i)}{1 - p_i^{L_{i, retry}+1}} \sum_{h=0}^j \frac{W_{i,h} - 1}{2} \quad (18)$$

$$E(B_i) = \frac{E(X_i)}{(1 - p_i)} p_i \quad (19)$$

$$E(N_{i, retry}) = \sum_{j=0}^{L_{i, retry}} \frac{j p_i^j (1 - p_i)}{1 - p_i^{L_{i, retry}+1}} \quad (20)$$

Let $D_i (i=0, \dots, N-1)$ denote the random variable representing the frame delay for the priority i class. Let T_o denote the time that a station has to wait when its frame transmission collides before sensing the channel again. Let $T_{ACK_timeout}$ and $T_{CTS_timeout}$ denote the duration of the ACK timeout and the duration of the CTS timeout, respectively. We have

$$E(D_i) = E(X_i) \delta + E(B_i) [p_s T_s + (p_b - p_s) T_c] \quad (21)$$

$$+ E(N_{i, retry}) (T_c + T_o) + T_s$$

$$T_o^{basic} = SIFS + T_{ACK_timeout} \quad (22)$$

$$T_o^{rts/cts} = SIFS + T_{CTS_timeout} \quad (23)$$

IV. CUSTOMIZING THE ANALYTIC MODEL

For the EDCF priority scheme, the number of access categories is 4, and we do not differentiate the window-increasing factor and the retry limit. Therefore, we have $N = 4$, and for the priority $i (i=0, \dots, 3)$ class, we have $\sigma_i = 2$, $L_{i, retry} = L_{Retry}$, and $W_{i,0} = CW_{min}[i]$. Assuming that there are n active stations in a Basic Service Set (BSS) and each station implements all 4 queues, there are total $4n$ queue entities, which are roughly equivalent to $4n$ stations in the original IEEE 802.11 DCF. More generally, let $n_i (i=0, \dots, 3)$ denote the number of active queues in the priority i class. We assume that any queue in any station always has frames ready to send. In order to use the general model, we also assume that one of the following two assumptions will be true.

- Each station belongs to one and only one priority class, and use only one queue that belongs to that class, **or**
- Each station has traffic of multiple priority classes. Each class traffic uses the queue belongs that class. If there is more than one queue finishing the backoff at the same time, we assume that the virtual collision handler will not make a choice about who will send, but it will let a collision occur in the wireless medium. With this assumption, each queue will be truly equivalent to a station in the original IEEE 802.11 DCF, and n_i , the number of active queues in the priority i class, can be treated as the number of active stations in the priority i class in the general analytical model.

V. NUMERICAL RESULTS

We conduct simulations to validate our results, and simulation results match analytical results very well. We omit

the results due to limited space. We use IEEE 802.11a as an example. IEEE 802.11a parameters can be found in [12, 13], as well as how to calculate $T_H + T_{E(L)}$ accurately [13]. The data rate is 6Mbps and the control rate is 6Mbps. The frame size is fixed as 1024 bytes. For demonstration purposes, we have. $N = 2$.

A. Effects of the initial window size

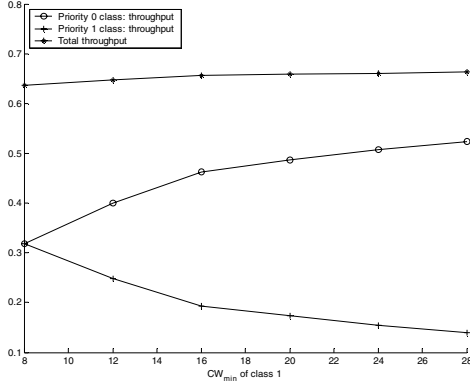


Fig. 2 throughputs (Differentiating CWmin)

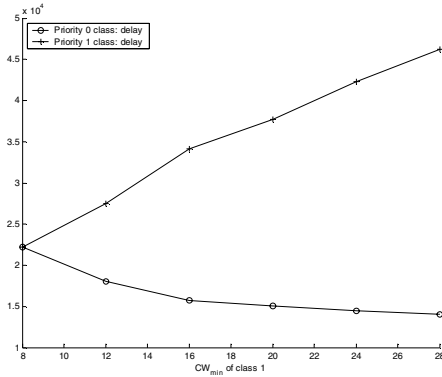


Fig. 3 Delay (μ sec.) (Differentiating CWmin)

Fig. 2 and Fig. 3 have the following parameters: $\sigma_0 = \sigma_1 = 2$, $L_{0, \text{retry}} = L_{1, \text{retry}} = 7$, $W_{0,0} = 8$, $CW_{0, \text{max}} = CW_{1, \text{max}} = 1024$ and $n_0 = n_1 = 10$. Fig. 2 (Fig. 3) shows saturation throughputs (saturation delays) over the minimum window size of class 1, $W_{1,0}$, which changes from 8 to 28. As illustrated in Fig. 2 (Fig. 3), when $W_{1,0} = 8$, saturation throughputs (saturation delays) are the same for both classes. As $W_{1,0}$ increases, the saturation throughput of class 1 decreases, the saturation throughput of class 0 increases, the saturation delay of class 0 decreases, and the saturation delay of class 1 increases. An interesting observation in Fig. 2 and Fig. 3 is that the throughputs/delays of class 0 and class 1 are symmetrical along a line parallel to the x-axis. This phenomenon indicates that class 0 can steal bandwidth from class 1 as the initial window size of class 1 ($W_{1,0}$) increases, whereas the total throughput of all classes increases a little. The reasons for this follow. As the initial window size of class 1 increases, stations in class 1 will delay accessing the channel so that the saturation throughput of class 1 will decrease and the saturation delay of class 1 will increase. Furthermore, the collision probability of class 0 will decrease so that the saturation throughput of class 0 will increase and the saturation delay of class 0 will decrease. Therefore, differentiating the initial window size has both the function of reducing collisions and the function of providing

priorities.

B. Effects of the backoff window-increasing factor

Fig. 4 and Fig. 5 have the following parameters: $\sigma_0 = 2$, $L_{0, \text{retry}} = L_{1, \text{retry}} = 7$, $W_{0,0} = W_{1,0} = 8$, $CW_{0, \text{max}} = CW_{1, \text{max}} = 1024$ and $n_0 = n_1 = 10$. Fig. 4 (Fig. 5) shows saturation throughputs (saturation delays) over the backoff window-increasing factor of class 1, σ_1 , which changes from 1.6 to 2.6. As illustrated in the figures, when $\sigma_1 < \sigma_0$, class 1 has better throughput and delay than class 0; when $\sigma_1 = \sigma_0 = 2$, they have the same throughput and delay; when $\sigma_1 > \sigma_0$, class 0 has better throughput and delay than class 1. Fig. 4 and Fig. 5 also has the same effect as in Fig. 2 and Fig. 3, respectively, that class 0 can steal bandwidth from class 1 with a similar reason, and the total throughput does not decrease. Differentiating the backoff window-increasing factor also has both the function of reducing collisions and the function of providing priorities.

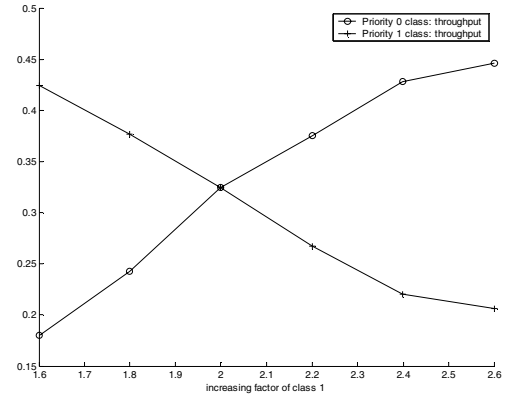


Fig. 4 Throughput by differentiating σ

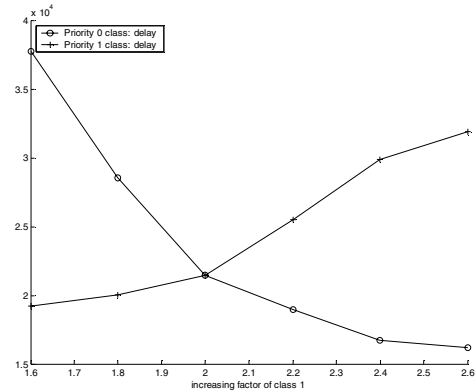


Fig. 5 Delay (μ sec.) by differentiating σ

C. Effects of the Retry Limit

Fig. 6, Fig. 7, and Fig. 8 have the following parameters: $\sigma_0 = \sigma_1 = 2$, $L_{0, \text{retry}} = 8$, $W_{0,0} = W_{1,0} = 16$, $CW_{0, \text{max}} = CW_{1, \text{max}} = 1024$ and $n_0 = n_1 = 10$. Fig. 6, Fig. 7, and Fig. 8 show saturation throughputs, saturation delays, and frame dropping probabilities, respectively, over the retry limit of class 1, $L_{1, \text{retry}}$, which changes from 0 to 8. As illustrated in the figures, class 1 has better throughput and delay than class 0 when $L_{1, \text{retry}} < L_{0, \text{retry}}$ and they have the same throughput and delay when $L_{1, \text{retry}} = L_{0, \text{retry}}$. Fig. 6 also has the same effect as Fig. 2

and Fig. 4 that class 0 can steal bandwidth from class 1 with a similar reason, and the total throughput does not decrease. Differentiating the retry limit also has both the function of reducing collisions and the function of providing priorities. Fig. 8 shows that class 1 has a higher frame dropping probability when $L_{1, \text{retry}} < L_{0, \text{retry}}$.

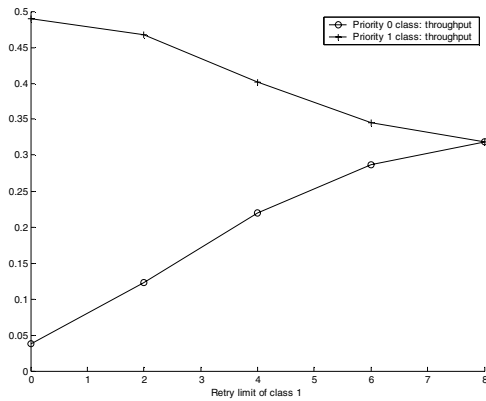


Fig. 6 Throughput by differentiating L_retry

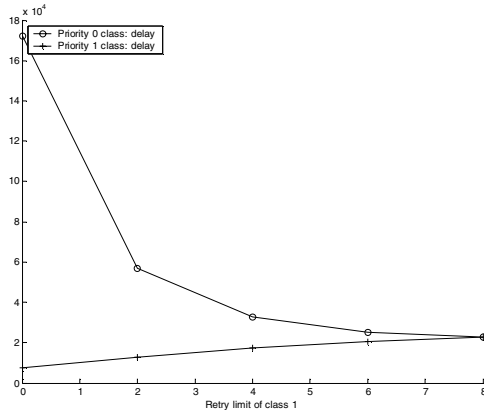


Fig. 7 Delay (μ sec.) by differentiating the retry limit

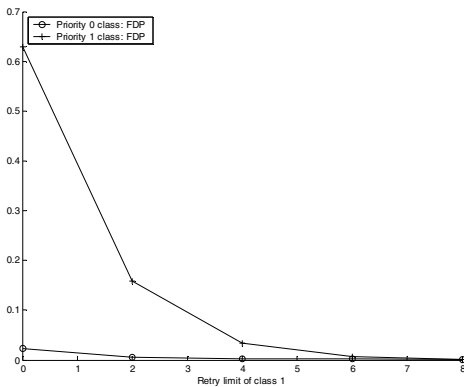


Fig. 8 Frame-dropping probability by differentiating the retry limit

VI. CONCLUSIONS

An improved analytical model was proposed to study the

emerging IEEE 802.11E EDCF in terms of saturation throughput, saturation delay, and frame dropping probability. Our study shows following results:

For all three backoff-based metrics (the initial window size, the retry limit, and the backoff window-increasing factor), one class can steal bandwidth from another if the later one increases the metric value, and the total throughput increases a little. This fact indicates that they are good metrics.

For real-time multimedia traffic with sensitive delay requirements, retransmitted frames may be too late to be useful, and a smaller retry limit is appropriate, whereas some non-real-time transmissions may need a larger retry limit to enhance reliable transmissions. Our results show that with a smaller retry limit, both throughput and delay can be improved with a higher frame dropping probability. This is good for real-time delay sensitive applications that can survive with frame loss, such as voice traffic.

A backoff-based metric has both the function of reducing collisions and the function of providing priorities.

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